



Answer any **five** questions of the following.

5 × 8 = 40

- ✓ (a) What is an Embedded System? Describe the Components and Constraints of an Embedded System. CLO3 4
- (b) Explain Four types of Sensor and Actuator with relevant examples. CLO3 4
- ✓ (a) Explain about processor technology and its classification. CLO3 4
- (b) Draw the architecture of Embedded System Hardware for Home Control Automation like Smart Home and briefly explain the key parts. CLO4 4
3. (a) What is Direct Memory Access (DMA)? Explain the working process of DMA. CLO3 4
- (b) Draw and Explain the block diagram of DMA chip 8257. CLO3 4
- ✓ 4. (a) Mention Types of Interrupt in 8086. CLO3 2
- (b) Explain Interrupt Execution with the help of Interrupt vector table in the 8086 microprocessors. CLO3 3
- (c) Briefly Mention Hardware Interrupt and Interrupt Priority in 8086. CLO3 3
- ✓ 5. (a) Draw the Pin Diagram of Microprocessor 8086 and Explain the followings: CLO3 4
- i. Address Status Line
- ii. NMI
- (b) Explain Horizontal and Vertical micro instruction. Mention two major differences of Hardwire Control Unit and Microprogrammed Control Unit. CLO3 4
- ✓ 6. (a) Draw the diagram for Micro Program Control Unit and Explain the Instruction Cycle for [ADD X<sub>1</sub>, X<sub>2</sub>]. CLO1 4
- (b) Analyze Wilkes Design for Micro Programmed CU for the following input sequence. CLO3 4
- i. 001
- ii. 111
- iii. 101
- iv. 011

7. (a) Mention the features of 8255 Programmable Peripheral Interface (PPI). CLO2 3
- (b) Draw the block diagram of 8255 PPI and describe the following: CLO2 5
- i. Data Bus Buffer
  - ii. Read Write Control Logic
  - iii. Group A and Group B Control
  - iv. Port A, Port B and Port C



# HAMDARD UNIVERSITY BANGLADESH

## Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 317: Computer Networks

Batch: 26<sup>th</sup>

Section: 1 & 2

Date: 08/09/2026

Final Examination

Semester: Spring 2026

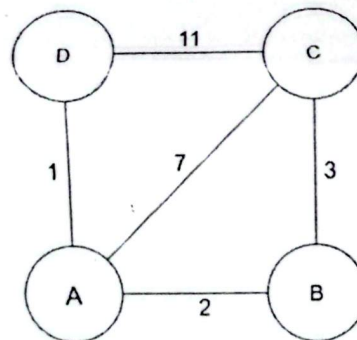
Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

- |                                                                                                                                                                    | CLOs | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 1 (a). Write the differences between Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).                                                         | CLO5 | 2     |
| (b). Define the <b>Count-to-Infinity</b> problem. Why does the <b>Count-to-Infinity</b> problem happen? Briefly explain the techniques for mitigating the problem. | CLO5 | 6     |
| 2 (a). Write the working procedure of the <b>Bellman-Ford</b> routing algorithm.                                                                                   | CLO5 | 3     |
| (b). Consider the given diagram:                                                                                                                                   | CLO5 | 5     |



Find the shortest path implementing the Distance Vector routing algorithm.

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|
| 3 (a). Define the Dynamic Host Configuration Protocol (DHCP). Illustrate the packet format of DHCP with brief explanation.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CLO5 | 3 |
| (b). Imagine you are a network administrator at a growing tech enterprise. On Monday morning, a remote software engineer named Ajim arrives at headquarters, enters a newly configured conference room, plugs an Ethernet cable into their laptop, and opens a web browser to join an all-hands video call. Within seconds, the laptop automatically configures its networking stack and seamlessly loads the company dashboard—without Alex ever touching the IP setting menu. Explain the complete end-to-end behind-the-scenes working procedure that makes this connection possible. | CLO5 | 5 |
| 4 (a). What is the Open Shortest Path First (OSPF)? Write the key characteristics of the OSPF protocol.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CLO5 | 4 |

(b). Explain the working procedures of the OSPF.

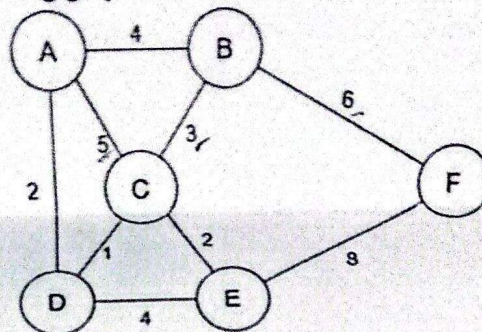
CLO5 4

(a). Define the Link State routing algorithm with its key features.

CLO5 2

(b). Consider the following graph:

CLO5 6



Construct the shortest path routing table of nodes A and F using the Link State routing algorithm.

6 (a). Define Hypertext Transfer Protocol (HTTP).

CLO1 1

(b). Write the differences between the Circuit switching and the Packet switching networks.

CLO1 3

(c). Explain the steps for getting IP address from domain name using Domain Name System (DNS) protocol with example.

CLO1 4

(a). Explain the significant messages of Internet Control Message Protocol.

CLO5 4

(b). Explain all kinds of addresses of Network Address Translation (NAT) using appropriate example and diagram.

CLO5 4



# HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering  
Department of Computer Science and Engineering  
Faculty of Science, Engineering and Technology

Course Code: MAT 317

Final Examination

Date: 9/13/2026

Course Title: Engineering Mathematics

Semester: Spring 2026

Total Marks: 40

Total Time: 2.00 Hours

4 × 10 = 40

Answer any four (04) out of the following questions:

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

| No.    | Questions                                                                                                        | CLOs         | Marks |
|--------|------------------------------------------------------------------------------------------------------------------|--------------|-------|
| 10 Q1. | (a) What do you mean by Fourier transformation?                                                                  | CLO1         | [1]   |
|        | (b) Show that the Fourier transformation of the following function exist. Also, calculate its Fourier transform. | CLO2<br>CLO3 | [5]   |
|        | $f(t) = \begin{cases} 0, & t < 0 \\ e^{-at}, & t > 0, a > 0 \end{cases}$                                         |              |       |
|        | (c) Find the Fourier transformation of the following function.                                                   | CLO3         | [4]   |
|        | $f(t) = e^{-a t } = \begin{cases} e^{at}, & t < 0 \\ e^{-at}, & t > 0 \end{cases}$                               |              |       |
| Q2.    | (a) What do you mean by inverse Fourier transformation?                                                          | CLO1         | [1]   |
|        | (b) Find the inverse Fourier transformation of the following functions:                                          | CLO3         | [5]   |
|        | (i) $f(\omega) = \frac{e^{4t\omega}}{3 + i\omega}$                                                               |              |       |
|        | (ii) $f(\omega) = \frac{e^{-i\omega}}{2(1 + i\omega)}$                                                           |              |       |
|        | (iii) $f(\omega) = \frac{e^{-2t\omega}}{2(4 + \omega^2)}$                                                        |              |       |
| Q2.    | (c) What do you mean by Sturm-Liouville problem? Check whether the following is Sturm-Liouville or not.          | CLO1<br>CLO3 | [4]   |
|        | $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + (x^2 + 1 + \lambda)y = 0; \quad y(0) = 0, y'(L) = 0$                     |              |       |
|        | Where $L > 1$ .                                                                                                  |              |       |
| Q3.    | (a) What do you mean by eigenvalue and eigen function?                                                           | CLO3         | [1]   |
|        | (b) Find the eigenvalue and eigen function of the following Sturm-Liouville problem.                             | CLO3         | [9]   |
|        | $\frac{d}{dx} \left( e^{2x} \frac{dy}{dx} \right) + (\lambda + 1)e^{2x}y = 0; \quad y(0) = 0, y'(\pi) = 0$       |              |       |
| Q4.    | (a) Write the Legendre polynomial of first kind.                                                                 | CLO1         | [1]   |
|        | (b) Prove the followings:                                                                                        | CLO2         | [9]   |
|        | (i) $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $m \neq n$                                                             |              |       |
|        | (ii) $\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$ if $m = n$                                                     |              |       |
|        | Where, $P_n(x)$ is the Legendre polynomial of first kind.                                                        |              |       |



- 10  
Q5.
- (a) Using the Rodrigues formula of Legendre polynomial, calculate the followings:  $P_0(x)$ ,  $P_1(x)$  and  $P_2(x)$ . CLO3 [3]
- (b) Show that  $(1 - 2xz + z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} z^n P_n(x)$  considering  $|x| \leq 1, |z| \leq 1$ . Where  $P_n(x)$  is the Legendre polynomial of first kind. CLO3 [7]
- Q6. (a) Find  $J_0(x)$  and  $J_1(x)$  using the definition of  $J_n(x)$ . CLO3 [2]
- (b) Prove the followings: CLO2 [4]
- (i)  $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$
- (ii)  $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$
- (c) If  $n$  is any integer then show that  $J_{-n}(x) = (-1)^n J_n(x)$  CLO3 [4]

**Faculty of Science, Engineering and Technology**  
**Department of Computer Science and Engineering**  
**CSE 313: Database Management System**

Batch: 26<sup>th</sup>

Semester: Spring 2026

Date: 03/09/2026

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings.

5×8= 40

1. (a) Write the differences between SQL and iSQL plus. CLO 2 3  
 (b) Show the differences between join and natural join in DBMS with example. CLO 4 5

2. Let us have the following Emp table:

| Emp_id | First_name | Last_name | Dept | Salary   | Job_ID | Dept_ID |
|--------|------------|-----------|------|----------|--------|---------|
| 31511  | Anik       | Das       | CSE  | 40,000/= | CS_TEA | 315     |
| 41211  | Susmita    | Roy       | EEE  | 37,000/= | EE_TEA | 412     |
| 22211  | Fahima     | Hossain   | ENG  | 48,000/= | EN_TEA | 222     |
| 53611  | Abul       | Hasnat    | Econ | 55,000/= | EC_TEA | 536     |
| 31512  | Shymol     | Roy       | CSE  | 66,000/= | CS_TEA | 315     |

- (a) Write the SQL to show the Last\_Name and dept of the employee whose salary is between 40,000/= and 60,000/=. CLO 3 2  
 (b) Write the SQL to show the First\_Name and dept of the employee who is neither in the CSE nor in the ENG department. CLO 3 2  
 (c) Write the SQL to show the First\_Name and dept of the employee whose salary is not in the 40,000/=, 37,000/= and 48,000/=. CLO 3 2  
 (d) Write the SQL to show the First\_Name and dept of the employee whose salary is more than 50,000/=. CLO 3 2

3. (a) Why 3NF is stronger than BCNF? CLO 4 2  
 (b) Consider the following Employees table: CLO 3 6

| Emp_id | First_name | Last_name | Email | Job_id | Salary | Department_id |
|--------|------------|-----------|-------|--------|--------|---------------|
|--------|------------|-----------|-------|--------|--------|---------------|

- i) Update employee 114's job\_id and salary to match that of employee 205 using subquery.  
 ii) Insert the following data into the above employees table using subquery:

| Emp_id | Last_name | Email         | Job_id   | Salary | Department_id |
|--------|-----------|---------------|----------|--------|---------------|
| 99999  | Taylor    | 123@gmail.com | St_clerk | 50000  | 50            |

4. (a) Write the types and guidelines for using sub queries in DBMS. CLO 4 3  
 (b) What is meant by a group function? Write the SQL to show the max, min, sum and average salary distinctly from the Emp table of Q(2). CLO 3 5

5. (a) State the key components of Trigger in DBMS. CLO 1 2  
 (b) Write the SQL to create a Trigger. CLO 3 2  
 (c) State the Guidelines and write the syntax for Modifying a Sequence. CLO 4 4

6. (a) What is meant by Cartesian products in DBMS? CLO 4 2  
 (b) State the rules of operator precedence used in SQL. CLO 2 2

- (c) Write the SQL to find the Cartesian product for the following **Emp** table of Q(2) CLO 3 4  
and the following **Dept** table

| Dept_ID | Dept | Dept_Location    | Dept_Contact |
|---------|------|------------------|--------------|
| 315     | CSE  | Science Building | 315-1212     |
| 412     | EEE  | Physics Building | 412-1212     |
| 222     | ENG  | Arts Building    | 222-1212     |
| 536     | Econ | Arts Building    | 536-1212     |

7. Consider the following **Student** table with variable repeating groups. Find 1NF, 2NF CLO 1 8  
and 3NF for the following Student table:

| Std_Id | Adm_Date | Std_name | Std_email    | Grade | Cr_Points | Course_Name  | Ob_Mark |
|--------|----------|----------|--------------|-------|-----------|--------------|---------|
| 31501  | 11/03/18 | Dev Joe  | dev@iewr.com | A     | 1         | Discrete     | 76      |
| 31501  | 11/03/18 | Dev Joe  | dev@iewr.com | A     | 1         | Algorithms   | 77      |
| 31501  | 11/03/18 | Dev Joe  | dev@iewr.com | A     | 1         | Fundamentals | 76      |
| 31502  | 12/03/18 | Dany Sam | dany@ter.com | B     | 4         | Algorithms   | 66      |
| 31502  | 12/03/18 | Dany Sam | dany@ter.com | B     | 4         | Fundamentals | 67      |



# HAMDARD UNIVERSITY BANGLADESH

## Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 311: Theory of Computing

Batch: 26<sup>th</sup>

Section: 2

Date: 01/09/2026

Final Examination

Semester: Spring 2026

Time: 2 Hours

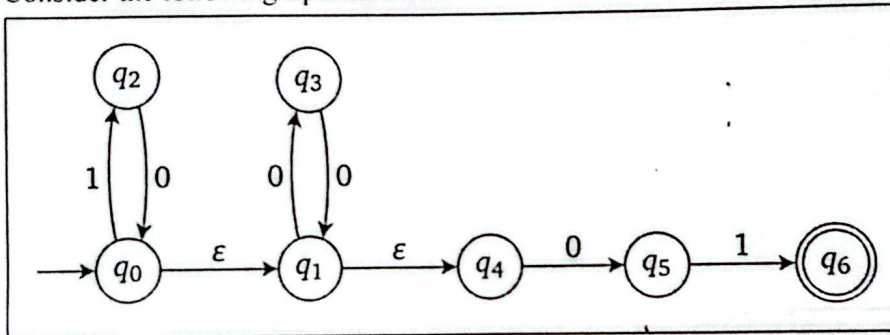
Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

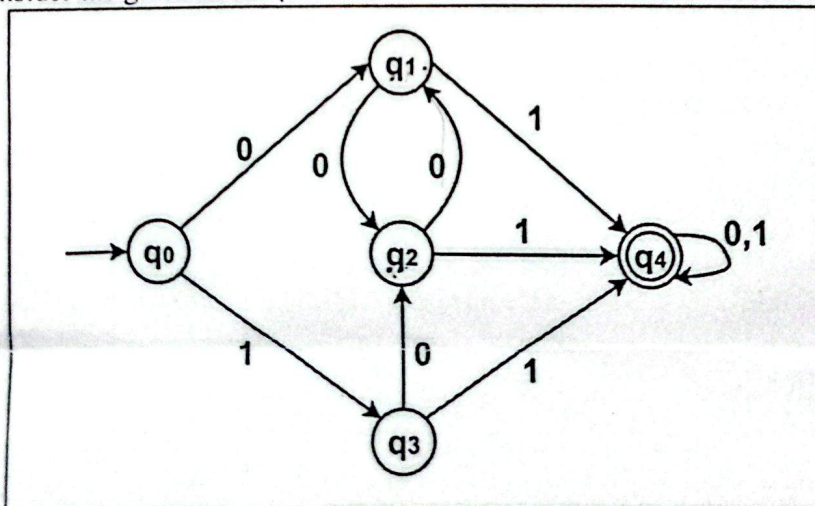
CLOs Marks

- 1 (a). Define the formal definition of the Epsilon Non Deterministic Finite Automata (NFA). CLO3 3
- (b). Consider the following Epsilon-NFA: CLO3 5



Convert the following Epsilon-NFA to an equivalent NFA.

- 2 (a). Explain the procedures for the Deterministic Finite Automata (DFA) minimization. CLO4 4
- (b). Consider the given DFA: CLO3 4



Minimize the given DFA.

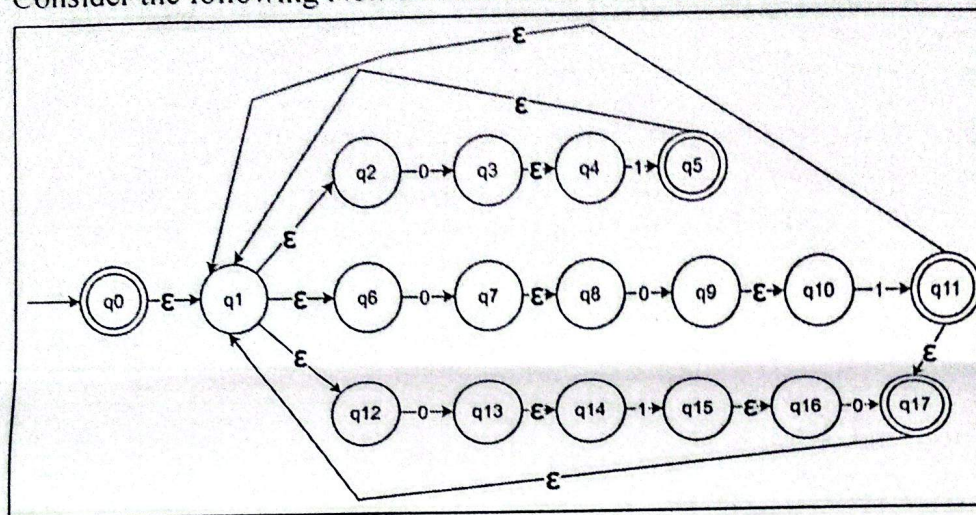
- 3 (a). Write the formal definition of a Moore machine with an example. CLO2 4
- (b). State the Mealy to Moore machine conversion algorithm. CLO4 4

4 (a). What is an NFA?

CLO3 2

(b). Consider the following Non-Deterministic Finite Automata (NFA):

CLO3 6



- Write the regular expression for the given NFA.
- Convert this NFA to an equivalent DFA.

5 (a). Define the **regular expression** in automata theory.

CLO1 2

(b). Write the operator precedence in regular expression.

CLO1 2

(c). Write the regular expression for the given regular languages:

CLO4 1×4=4

- $\{w \mid w \text{ starts with } 0 \text{ and has odd length, or starts with } 1 \text{ and has even length}\}$
- $\{w \mid w \text{ begins with a } 1 \text{ and ends with a } 0\}$
- Starts with 'bab', followed by zero or any number of a's or b's
- $\{w \mid w \text{ contains at least three } 1\text{'s}\}$

6 (a). Define the Context-Free Grammar (CFG).

CLO2 2

(b). Write the CFG rules for the following expression:

CLO4 2

$$\{0^n 1^n \mid n \geq 0\} \cup \{1^n 0^n \mid n \geq 0\}$$

(c). Consider the following CFG:

CLO4 2×2=4

$$\langle \text{EXPR} \rangle \rightarrow \langle \text{EXPR} \rangle + \langle \text{TERM} \rangle \mid \langle \text{TERM} \rangle$$

$$\langle \text{TERM} \rangle \rightarrow \langle \text{TERM} \rangle \times \langle \text{FACTOR} \rangle \mid \langle \text{FACTOR} \rangle$$

$$\langle \text{FACTOR} \rangle \rightarrow ( \langle \text{EXPR} \rangle ) \mid a$$

Prove the given expressions:

- $a + a \times a$
- $(a + a) \times a$

7 (a). Define the formal definition of a Push Down Automata (PDA).

CLO2 2

(b). Convert the following CFG into an equivalent CFG in Chomsky normal form:

CLO4 6

$$A \rightarrow BAB \mid B \mid \epsilon$$

$$B \rightarrow 00 \mid \epsilon$$